

Case Study: Enterprise AI-Powered Lead Qualification & Multi-Channel Routing Pipeline

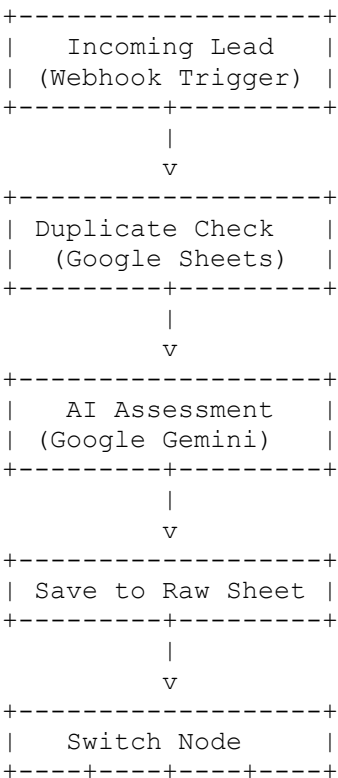
Executive Summary

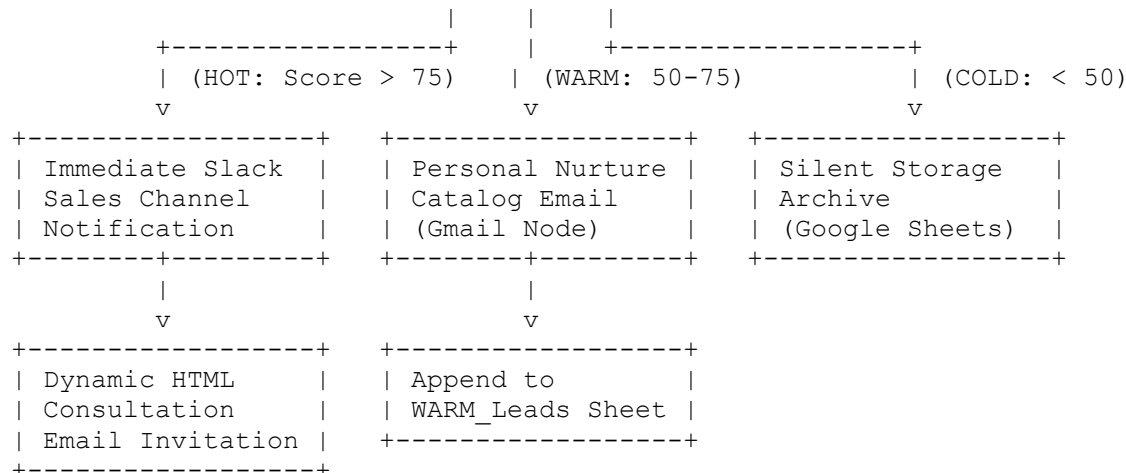
Real estate agencies in high-growth luxury markets face immense friction during client acquisition. High inquiry volumes, mixed lead intent, and manual processing delay reaction times, leading to missed deals and over-burdened sales teams.

To address this challenge, an automated, multi-channel lead qualification and distribution engine was engineered using **n8n**, **Google Gemini AI**, **Slack**, and **Google Sheets**. The solution ingests raw lead Webhooks, evaluates purchase intent in real time, routes leads across distinct operational tiers (**HOT**, **WARM**, **COLD**), executes personalized email outreach, and maintains a structured archive.

Technical Architecture & Data Flow

The architecture operates on an event-driven flow, ensuring zero human bottleneck from lead capture to initial sales alert.





Core System Functionality

1. Ingestion & Pre-Processing

- **Webhook Endpoint:** Ingests JSON payloads containing contact details, location preference, property type, budget, and timeline.
- **Deduplication Check:** Queries existing records to avoid re-entry or double-messaging existing prospects.

2. AI Lead Qualification Engine

- **Model:** Google Gemini via structured JSON output prompt engineering.
- **Scoring Logic:** Evaluates timeline urgency, target budget, preferred location demand, and purchase intent.
- **Outputs:** Returns a structured response containing AI Score (0–100), Category (HOT, WARM, COLD), and concise AI Reasoning.

3. Multi-Tier Branching Strategy

Branch Tier	AI Score Range	Key Triggers & Operational Workflow
HOT	\$76 - 100\$	Instant High-Intent Response: Sends immediate real-time alert to Slack sales channel. Dispatches a custom inline-styled HTML invitation email with direct calendar scheduling options.

Branch Tier	AI Score Range	Key Triggers & Operational Workflow
WARM	\$50 - 75\$	Mid-Funnel Nurturing: Automatically emails a curated property catalog and market guide. Logs lead details into a dedicated <code>WARM_Leads</code> Google Sheet tab with a <code>Pending</code> follow-up status.
COLD	\$0 - 49\$	Silent Archival: Stores full record silently in a <code>COLD_Leads</code> database archive tab for future automated email broadcasts or periodic audits, bypassing sales alerts.

Technical Features & Optimization Highlights

- 1. Inline-Styled HTML Email Delivery:**
 - Designed customized, responsive HTML templates for luxury branding.
 - Suppressed native n8n branding attributions and configured fallback rendering for strict corporate email clients.
- 2. Fault-Tolerant Dynamic Mapping:**
 - Engineered fallback expressions (e.g., `={{ \ (json['Email'] || \) json['email'] }}`) across JavaScript and sheet nodes to ensure field mapping integrity despite varied incoming payload structures.
- 3. Data Security & Privacy Integrity:**
 - Automated secure handling of client contact parameters and isolated sales notifications to authorized Slack workspace channels only.

Impact & Business Metrics

- Response Time:** Decreased average response time for HOT leads from **hours to under 2 seconds**.
- Sales Focus:** Reduced noise for sales agents by filtering out COLD inquiries and keeping non-urgent leads in dedicated nurturing queues.
- Data Accuracy:** Consolidated 100% of incoming inbound lead metrics into structured tracking environments without manual data-entry overhead.